

Clinical Risk Assessment Report: Patient 60

1. Primary Outcome & Risk Summary

- Target: Occult Lymph Node (OLN) Metastasis
- Model Prediction: Negative for Metastasis
- Prediction Confidence: Moderate Confidence
- Absolute Predicted Risk: 10.2% (Diagnostic Threshold: 22.1%)
- Relative Risk Multiplier: 0.46x the diagnostic threshold

Clinical Takeaway:

The machine learning model classified this patient as Low-Risk for Occult Lymph Node (OLN) Metastasis. The primary driver determining this prediction is the Small Zone Emphasis (PET) (GLZLM_SZE.PET), which reduced the patient's absolute risk by 2.8%. Biologically, this feature reflects the proportion of small, fine texture zones, often correlating with a highly chaotic and erratic micro-architecture.

2. Key Pathological & Imaging Drivers (Elevating Risk)

No major risk-elevating features observed in the top drivers.

3. Protective / Mitigating Factors (Reducing Risk)

Small Zone Emphasis (PET) (GLZLM_SZE.PET) Value: -0.32 [Avg (32th %ile)] (-2.8%)

Reflects the proportion of small, fine texture zones, often correlating with a highly chaotic and erratic micro-architecture.

Large Zone Emphasis (CT) (GLZLM_LGZE.CT) Value: -0.18 [Avg (54th %ile)] (-1.4%)

Reflects the presence of large homogeneous areas; lower values suggest a chaotic, heterogeneous microenvironment.

Long Run High Gray-Level Emphasis (PET) (GLRLM_LRHGE.PET) Value: -0.25 [Avg (54th %ile)] (-0.6%)

Points to continuous, elongated bands of high-density or highly active tumor tissue.

75th Percentile Tumor Density (CONVENTIONAL_HUQ3) Value: 0.34 [Avg (53th %ile)] (-0.6%)

Reflects the overall physical density, solid components, and general attenuation characteristics of the tumor.

PET Radiomic Metabolic Feature (GLZLM_ZLNU.PET) Value: -0.4 [Bottom <1%] (-0.5%)

Metabolic PET feature reflecting glucose avidity and cellular metabolic rate.

Short Run Low Gray-Level Emphasis (CT) (GLRLM_SRLGE.CT) Value: -0.14 [Avg (47th %ile)] (-0.5%)

Indicates fragmented areas of low density, often associated with diffuse micro-necrosis or loose tissue structure.

Tumor Local Contrast (PET) (GLCM_Contrast(=Variance).PET) Value: -0.57 [Avg (27th %ile)] (-0.4%)

Measures local variations and sharp transitions in density or metabolism, signifying an unstructured and variable tumor matrix.

Minimum Tumor Density (CONVENTIONAL_HUmin) Value: -0.44 [Avg (31th %ile)] (-0.4%)

Reflects the least dense areas within the tumor on CT, often corresponding to cavitation, necrosis, or cystic degeneration.

Machine Learning Feature Attribution (SHAP Decision Path)

